Dijie Zhu

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Research Interests

My research interests are foundation model training, policy evaluation and runtime correction. I develop training schedulers, policy evaluators, and anomaly detectors that learn from limited, noisy, or biased data and scale with growing datasets.

Education

University of California, Los Angeles — Sep 2024 – Jun 2026
Master of Science — *Los Angeles, CA, USA*

- Electrical and Computer Engineering major
- GPA: 3.822/4.0

University of Electronic Science and Technology of China & University of Glasgow — Sep 2020 – Jun 2024
Chengdu, China
Bachelor of Engineering with First Class Honours

- Electrical and Electronic Engineering major
- GPA: 3.88/4.0 Rank: 3/169

Selected Publications

SCAPE: Scenario-Conditioned Simulation-Augmented Policy Evaluation
Dijie Zhu, Seunghun Oh, Ruopeng Huang, Zhiyu Huang, Jiaqi Ma, Chen Tang
Conference on Robot Learning (CoRL), 2026. [arXiv]

Earlier publications in wireless communications are available on Google Scholar.

Research Experience

Georgia Tech TASL | Research Assistant — Atlanta, GA
Supervisor: Prof. Jiachen Li — Sep 2026 – Present

- Investigating **uncertainty estimation and runtime recovery** for long-horizon vision-language-action (VLA) policies.
- Studying runtime evaluators that assess action reliability and identify potential failures during policy execution.
- Exploring **evaluator-assisted decision making** to compare candidate actions and determine when replanning, correction, or recovery is necessary.

UCLA TAMS Lab | Research Assistant — Los Angeles, CA
Supervisor: Prof. Chen Tang — Aug 2025 – Jun 2026

- Developed **SCAPE**, a scenario-conditioned framework that combines limited paired target/surrogate evaluations with large-scale simulation rollouts to predict real-world policy performance on individual scenarios.
- Designed a two-stage pipeline to correct scenario-dependent simulation bias, augment training with target-aligned pseudo-labels, and produce calibrated 95% uncertainty intervals through conformal prediction.
- Evaluated SCAPE on autonomous driving in nuPlan and quadruped locomotion across Sim2Sim and Sim2Real settings, including velocity tracking on a physical Unitree Go2.
- Reduced prediction error by up to **34.7%** on nuPlan, **27.7%** on Go2 Sim2Sim, and **11.3%** on Go2 Sim2Real, with **10–60% paired-label savings**.

UCLA Information Theory and Systems Lab | Research Assistant — Los Angeles, CA
Supervisor: Prof. Suhas Diggavi — Mar 2025 – Nov 2025

- Analyzed and reproduced FixMatch, FreeMatch, and InfoMatch components in PyTorch, including confidence filtering, dynamic thresholds, weak/strong augmentation consistency, EMA evaluation, and multi-seed benchmark pipelines.
- Designed a FixMatch–FreeMatch interpolation strategy with consistency-driven

threshold scheduling to reduce early-stage pseudo-label noise while improving late-stage utilization of unlabeled samples.

- Improved test error relative to strong baselines by **4.65%** on CIFAR-10 with 10 labels, **3.82%** on CIFAR-10 with 4000 labels, and **0.82%** on CIFAR-100 with 10k labels; reached 90% of final performance in **57** vs. **110** epochs for the baseline.

Selected Projects

NN-Aided Digital Self-Interference Cancellation under Time-Varying Channels — Los Angeles, CA, Sep 2024 – Jan 2025

Neural Networks, Signal Processing, Full-Duplex Communications

- Developed an MLP-based digital self-interference canceller with linear memory-polynomial feature preprocessing to adapt to time-varying channels without frame-wise neural-network retraining.
- Achieved a **9.81 dB** mean cancellation gain and **93.1%** lower frame-to-frame variance than adaptive memory-polynomial cancellation with 200 pilot symbols; also outperformed standard MLP and residual-NN baselines.

Human-Like NPC Pathfinding with Reward-Guided RRT — Los Angeles, CA, Jan 2025 – Mar 2025

Motion Planning, RRT, Trajectory Evaluation

- Developed a reward-guided RRT planner in a custom Unity environment, incorporating goal bias, obstacle and road awareness, spline smoothing, redundant-node removal, and speed alignment to generate human-like NPC trajectories.
- Benchmarked baseline, reward-guided, and post-processed RRT against 300 human trajectories from 10 participants using DTW and speed-consistency metrics; achieved the lowest trajectory mismatch and reduced speed deviation by **64.9%** versus baseline RRT.

Autonomous Mobile Robot: Multi-Environment Task Fulfillment — Chengdu, China, Mar 2023 – Jun 2023

Embedded Systems, Control, Perception

- Led a **10**-person team to build an autonomous mobile robot integrating mechanical design, sensing, control, and wireless communication under a **1000 RMB** budget.
- Designed and integrated the HC-12 wireless communication module, PID controllers, and embedded control software for autonomous task execution.
- Validated line following, obstacle avoidance, ramp traversal, wireless command execution, and ball delivery, achieving **98%** average task completion and a final score of **92/100**.

Technical Stack

Research Areas: Policy evaluation, RL, Sim2Real, robotics, autonomous driving, VLA, computer vision, semi-supervised learning, uncertainty quantification, conformal prediction, OOD generalization

Programming / ML: Python, C/C++, MATLAB, PyTorch, TensorFlow, Jupyter

Robotics / Simulation: Isaac Sim, MuJoCo, Gymnasium, Unitree Go2, ROS, Intel RealSense cameras

Tools: Linux, Bash, Git, Docker, Hydra, Ray, Weights & Biases, Nvidia Jetson